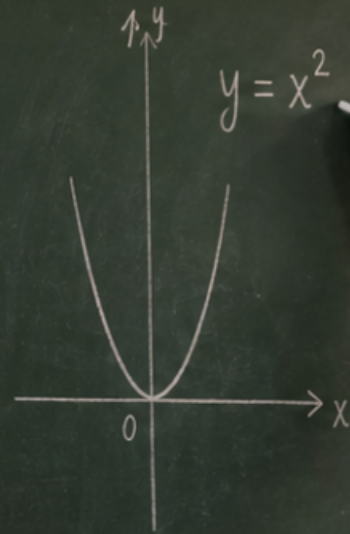


CHALKLINE

The board
stays with you.

Live blackboard lessons.
Built for weak connections.



YARRA VIVEK / INDIVIDUAL ENTRY

Snapdragon AI Lab Build & Present Challenge

PRODUCT PROPOSAL • AI-GENERATED CONCEPT VISUAL

THE PROBLEM

A missed symbol. A lost explanation.

A small minus sign matters more than a smooth video frame.

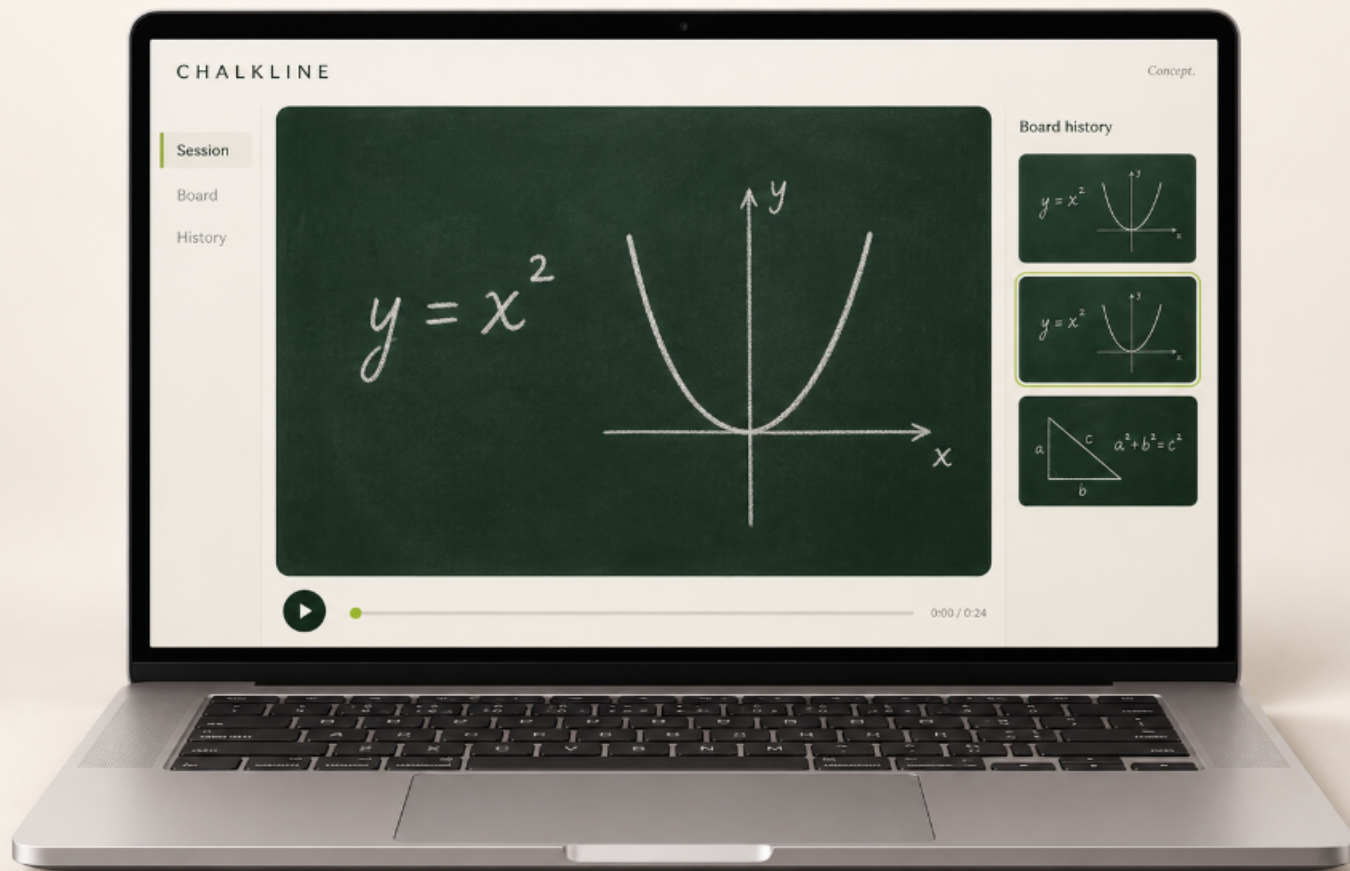
When the connection stalls, a student can lose the exact step the teacher has already erased.

Keep the teaching style.
Change how the board travels.



Illustrative learner scenario; concept image.

A live board that remembers.



Read

Original handwriting.
No retyped equations.

Rewind

Return to an earlier
board state.

Recover

Catch up after
a connection drop.

64 kbps

**For the board, the voice,
and the transport.**

Proposed average allocation

24 Audio

32 Board

8 Overhead

All values in kbps. Budget hypothesis;
bursts and recovery must fit the cap.

DESIGN TARGET — NOT A MEASURED RESULT

Spend compute beside the teacher. Save bandwidth for the learner.

01 Observe

USB camera → board crop
Perspective alignment on CPU.

02 Separate

NPU segmentation identifies
the presenter and occlusion.

03 Preserve

CPU tracks observed pixels,
erasures and board versions.

04 Deliver

Changed regions + audio.
A browser rebuilds the board.

LOCAL AI

Snapdragon X
HP PC

Candidate model:
DeepLabV3+MobileNet
via ONNX Runtime QNN

Treat the board as a document that changes over time.

PRESERVE WHAT WAS SEEN

Hold the last observed board pixels.
Mark hidden or stale regions.
Never invent missing handwriting.

RESPECT WHAT WAS ERASED

Confirm erasures across clear frames.
Keep earlier versions available.

The hardest failure

A teacher covers a line.
The system mistakes it
for an erasure.

Conservative updates
and visible freshness
are part of the product.

One lesson. Three delivery methods. The same constrained connection.

CHALKLINE

Board changes + audio

Tuned video

Low frame rate + audio

Snapshots

Periodic images + audio

WRITE → BLOCK → ERASE → DISCONNECT → REJOIN

Readability

Blind symbol
transcription

Continuity

Audio intelligibility
and board delay

Recovery

Correct board after
a 10-second drop

Efficiency

Total bytes and
CPU/NPU profile

A FOCUSED FIRST RELEASE

One teacher laptop. Any student browser.

Start with independent tutors and college lecturers who already teach from a physical board.

BUILD

One camera. One board.
A ten-minute lesson.

PROVE

Readable symbols.
Correct recovery.

PILOT

One teacher and
5–10 adult learners.

CHALKLINE

The next line should reach everyone.

Evidence behind the proposal

- | | |
|----------------------------------|---|
| 1 Qualcomm AI Hub | DeepLabV3-Plus-MobileNet model catalog |
| 2 ONNX Runtime | QNN execution provider for Qualcomm hardware |
| 3 Microsoft Teams Support | Camera-based board enhancement; blackboards currently unsupported |
| 4 Microsoft Learn | Teams network preparation and bandwidth guidance |
| 5 Challenge listing | Snapdragon AI Lab Build & Present Challenge |

Sources reviewed 8 September 2026. Full clickable references are in speaker notes and the project brief.

Concept-stage submission. All product imagery is AI-generated; no pilot, benchmark or deployment is claimed.